Physics Potential
LHμC@3.7 TeV & FCC- μh@10 TeV

$p(7000\ \mathrm{GeV}) + \mu^+(500\ \mathrm{GeV})$ $p(50000\ \mathrm{GeV}) + \mu^+(500\ \mathrm{GeV})$

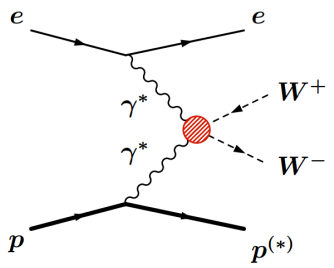
Hamzeh Khanpour

AGH University of Kraków

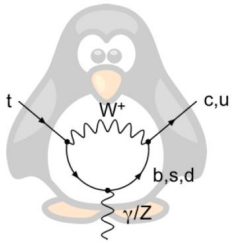
e/mu-LHC proposal: initial discussion; Dec 9, 2025, CERN

Outline: LHμC@3.7 TeV & FCC- μh@10 TeV

- Cross sections for wide spectrum of $\gamma\gamma$ processes : τ-pairs, W^+W^-, ZZ, higgsinos, sleptons, $t\bar{t}$



- Anomalous quartic gauge couplings : aQGC@$\gamma\gamma W^+W^-$



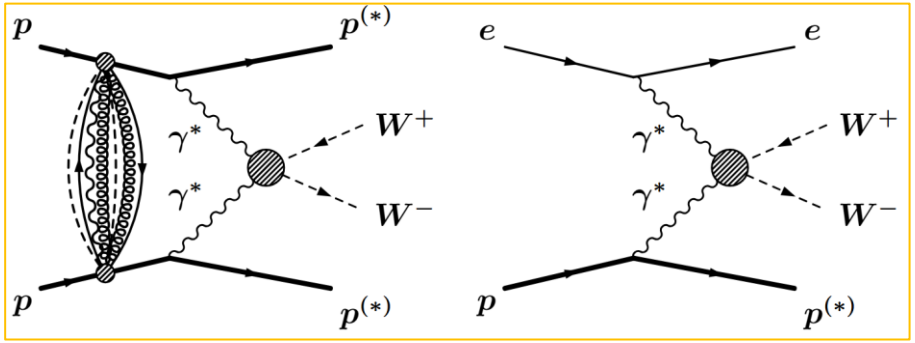
- Top-quark flavor-changing neutral currents : top-FCNCs

Cross sections for wide spectrum of $\gamma\gamma$ processes

- Comprehensive survey of studies of high energy photon-photon interactions at the LHμC@3.7 TeV & FCC- μh@10 TeV, for the photon-photon center-of-mass energy of up to 3 TeV.

- Wide spectrum of $\gamma\gamma$ processes could be studied, including, in particular, the exclusive production of lepton pairs, Higgs boson, W and Z bosons, $t\bar{t}$, as well as pairs of charged supersymmetric particles.

- As we will see, very high statistics of these processes are expected to be achieved at the LHμC@3.7 TeV & FCC-μh@10 TeV.

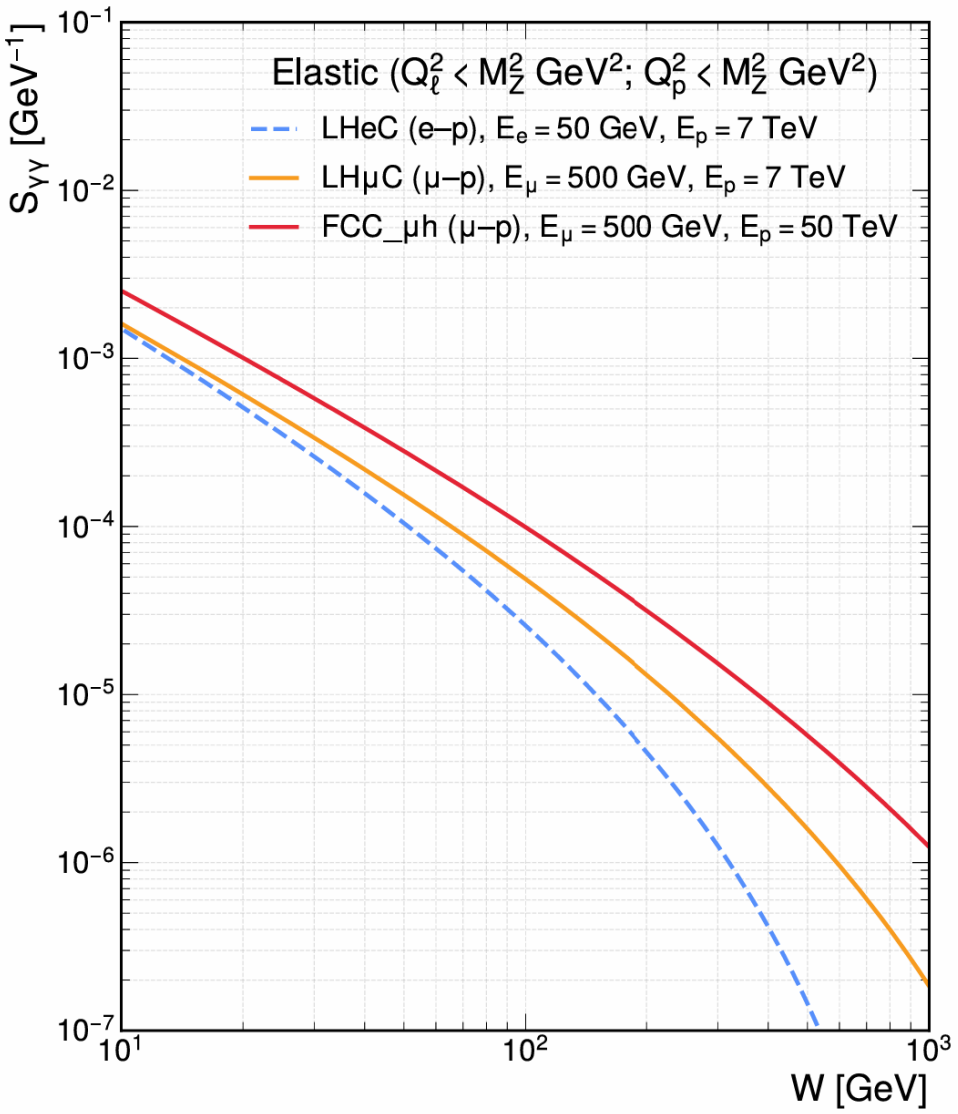


Feynman diagrams representing the exclusive $W^{\pm}$ boson pair production via photon-photon fusion at the LHC (left) and future collider LHeC (right).

Luminosity Spectrum $S_{\gamma\gamma}$

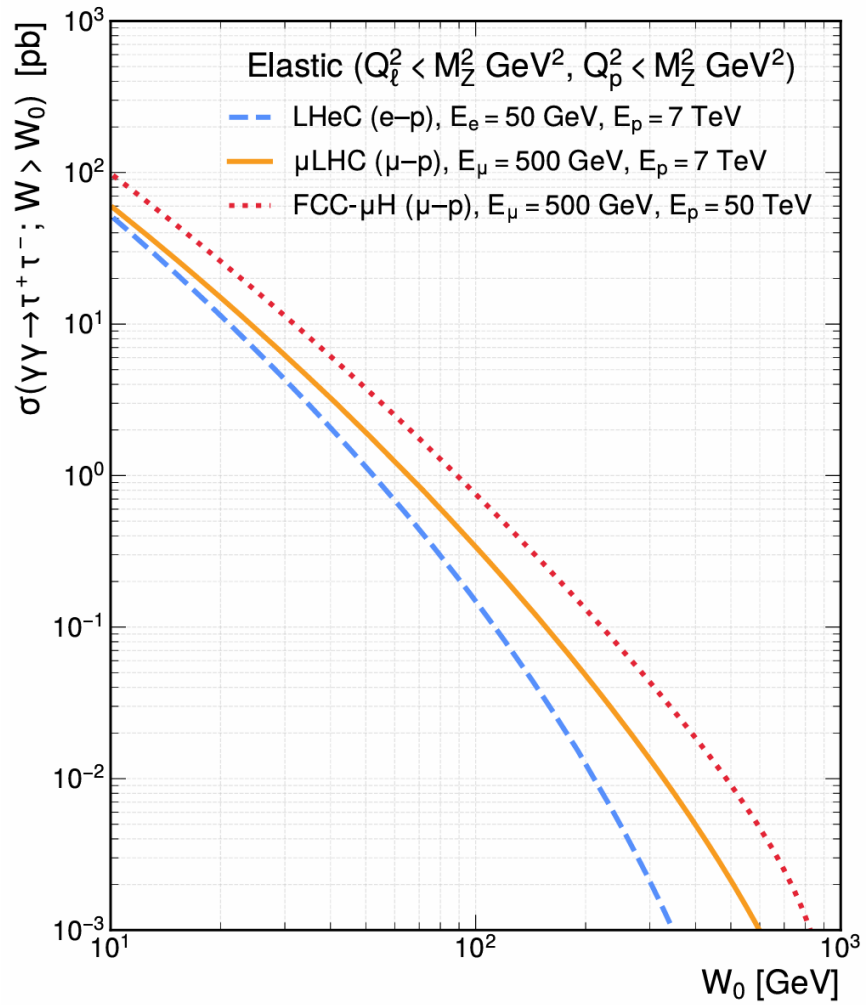
- Elastic luminosity spectrum $S_{\gamma\gamma}$ at the LHeC@1.2 TeV, LHμC@3.7 TeV & FCC-μh@10 TeV.

- It represents the fraction of the μp luminosity available for $\gamma\gamma$ collisions.



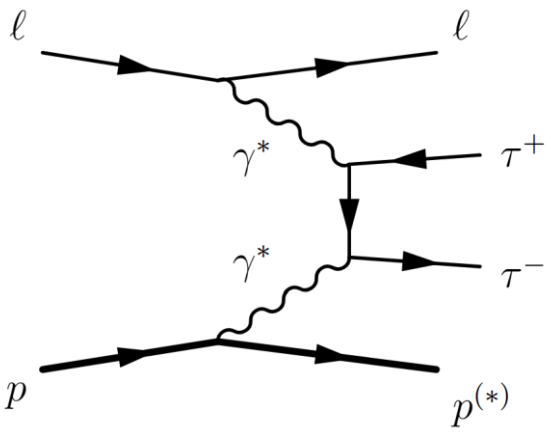
Elastic luminosity spectrum $S_{\gamma\gamma}$ at the LHeC@1.2 TeV, LHμC@3.7 TeV & FCC- μh@10 TeV.

Photon-photon factory of τ pairs



Elastic τ+τ− production cross sections (in pb) at the LHeC@1.2 TeV, LHμC@3.7 TeV & FCC- μh@10 TeV.

σ^{el}(LHeC@1.2 TeV) = 50.37 pb
σ^{el}(μLHC@3.7 TeV) = 58.56 pb
σ^{el}(FCC- μh@10 TeV) = 95.67 pb

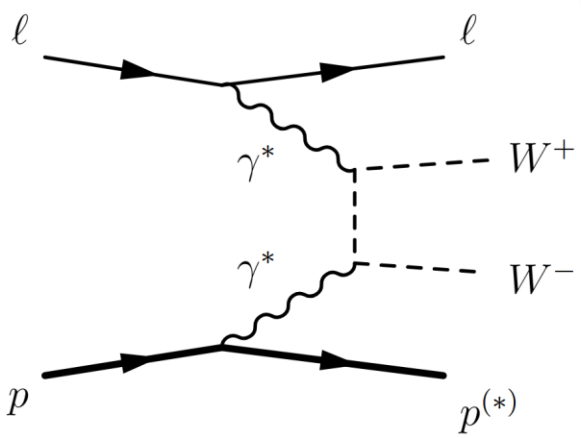


Feynman diagrams representing the exclusive production of τ pairs via photon-photon fusion at a lepton-hadron collider.

Very high statistics of these processes are achieved at the LHeC@1.2 TeV, LHμC@3.7 TeV & FCC- μh@10 TeV.

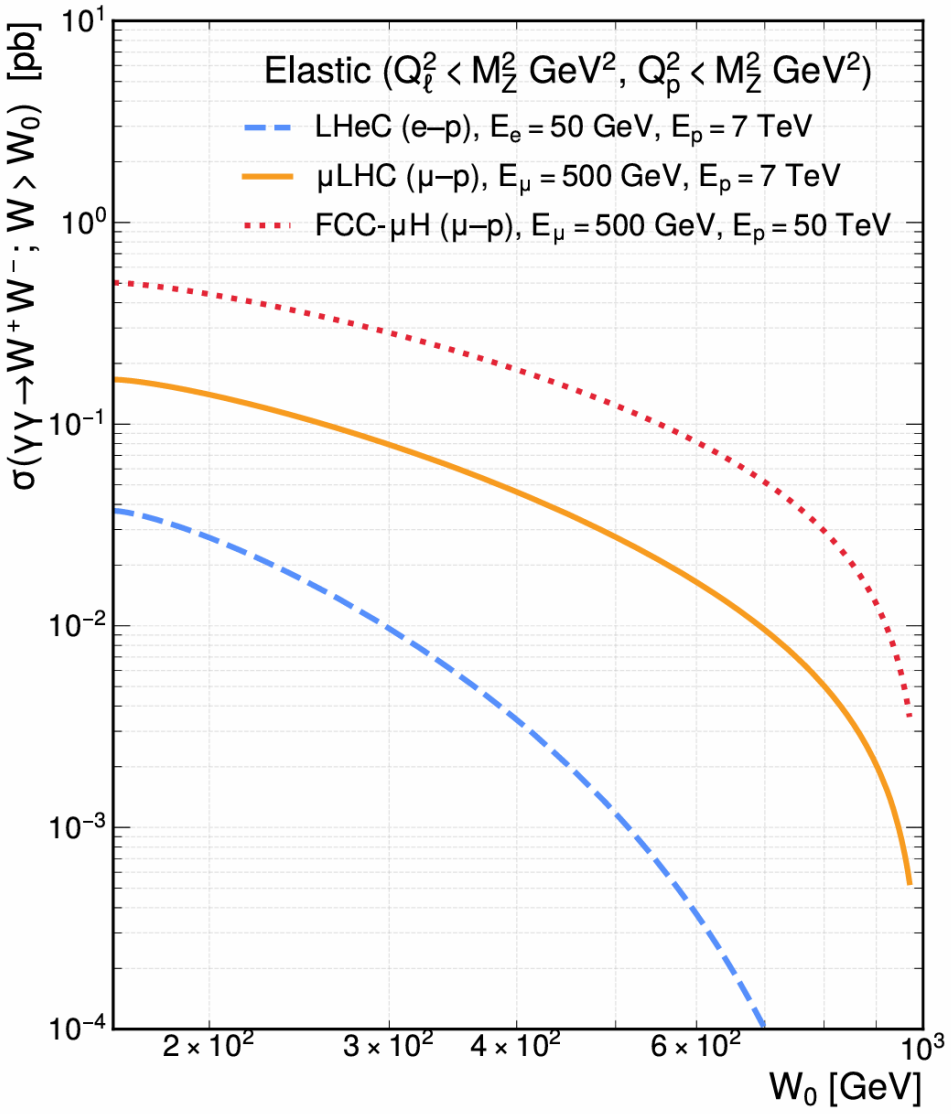
FCC- μh has the largest yield and hardest tail, hence the strongest statistical reach.

Exclusive production of W-boson pairs



Feynman diagrams representing the exclusive production of W-boson pairs via photon-photon fusion at a lepton-hadron collider.

σ^{el}(LHeC@1.2 TeV) = 0.037 pb
σ^{el}(μLHC@3.7 TeV) = 0.166 pb
σ^{el}(FCC- μh@10 TeV) = 0.503 pb



Elastic $W^\pm$ production cross sections (in pb) at the LHeC@1.2 TeV, LHμC@3.7 TeV & FCC- μh@10 TeV.

Unique Z boson laboratory

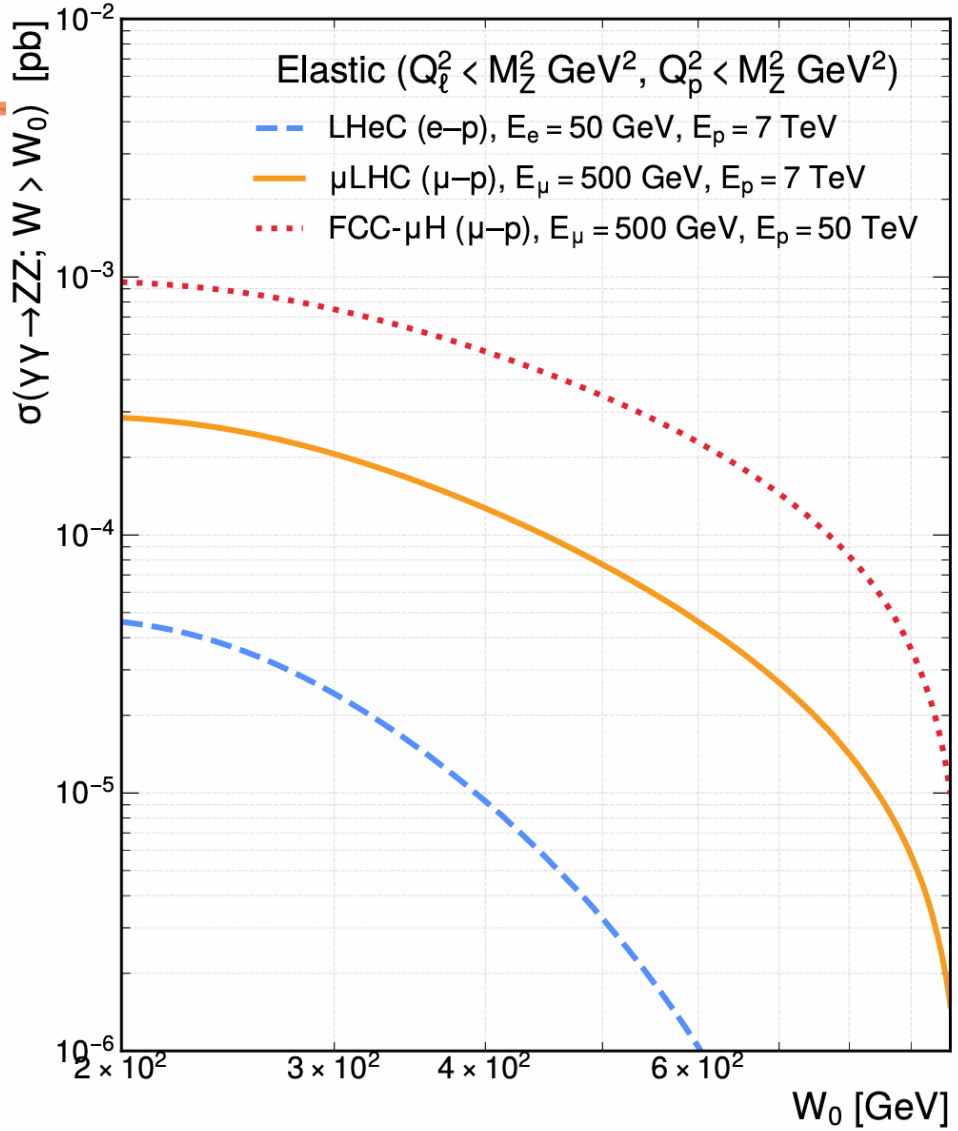
- In SM, $\gamma\gamma \to ZZ$ not allowed at tree level, lepton-hadron's $\gamma\gamma \to ZZ$ is a pure one-loop SM process.

- Higher mass tails of M_{ZZ} at the μLHC & FCC-μh will show more sensitivity to the nQGCs ($ZZ\gamma\gamma$).

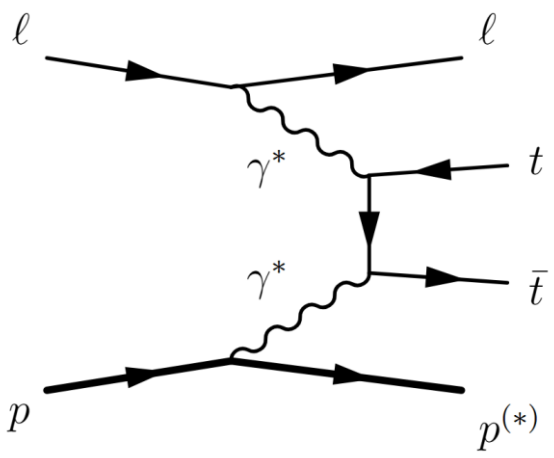
- FCC-μh (and μLHC) uniquely would accesses the ZZ high-mass regime, turning μp into a precision Z-boson laboratory.

σ^{el}(LHeC@1.2 TeV)	=	0.046 fb
σ^{el}(μLHC@3.7 TeV)	=	0.284 fb
σ^{el}(FCC- μh@10 TeV)	=	0.955 fb

Elastic ZZ production cross sections (in pb) at the LHeC@1.2 TeV, LHμC@3.7 TeV & FCC- μh@10 TeV.



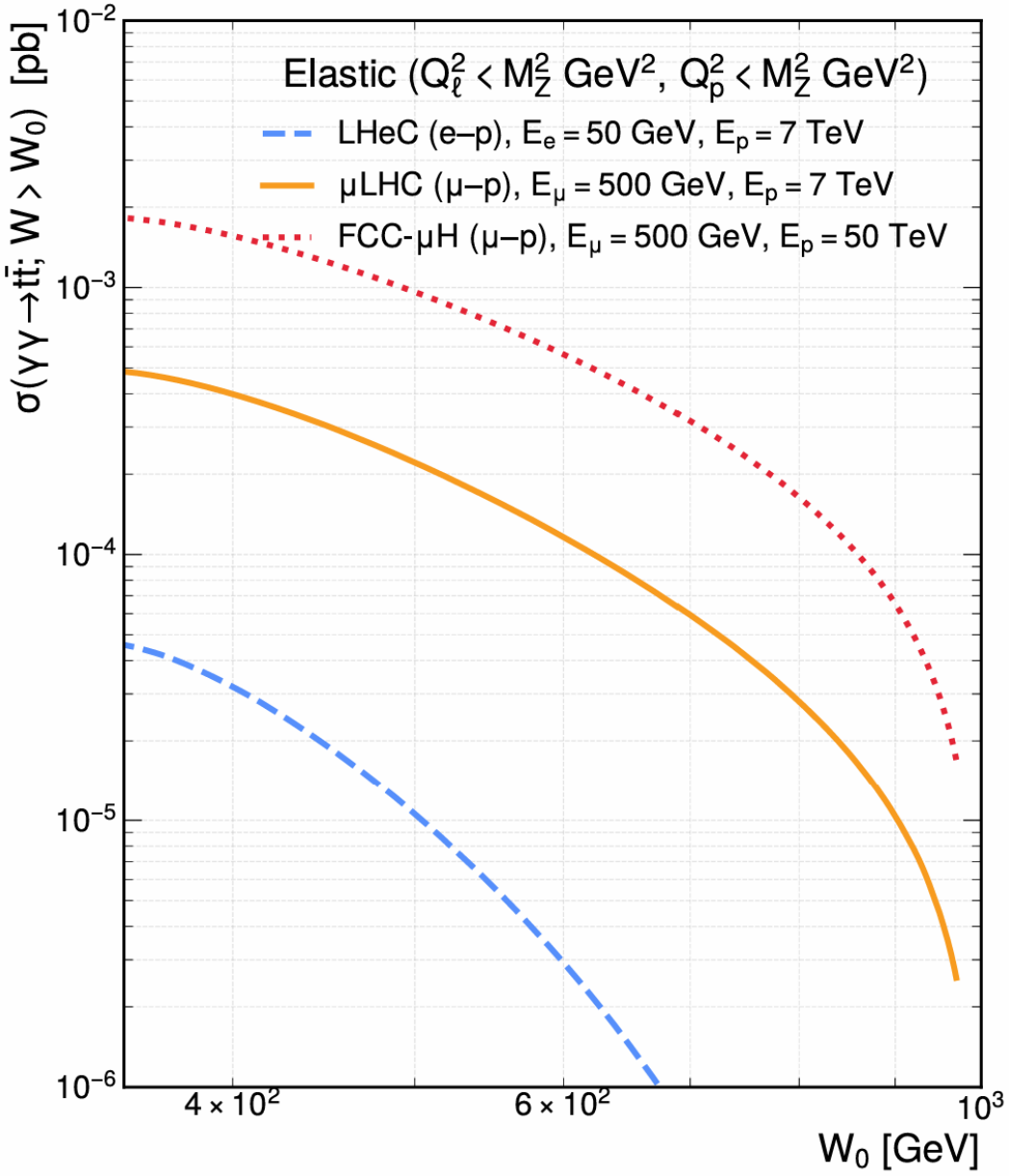
Exclusive production of $t\bar{t}$



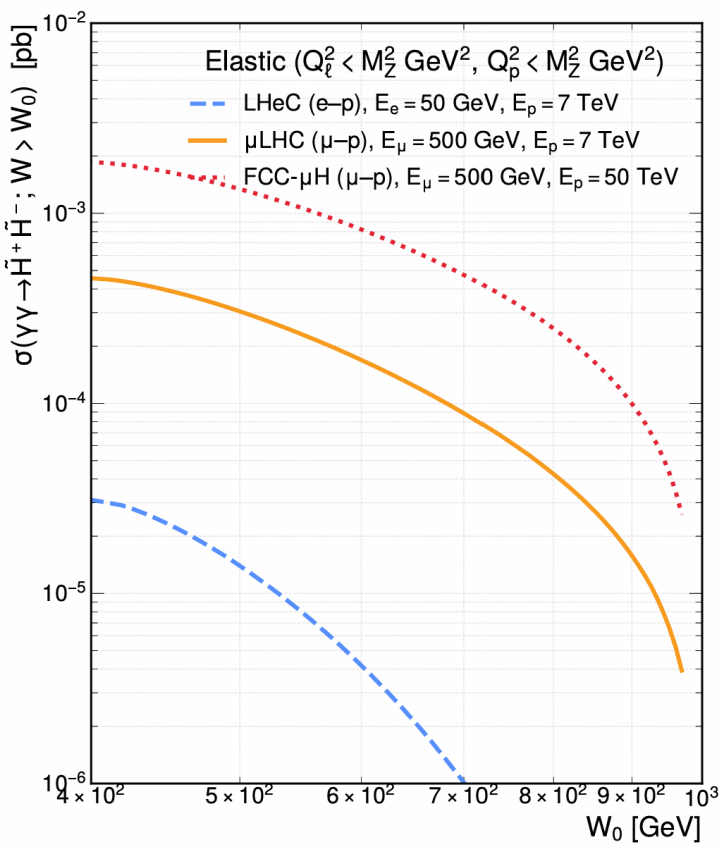
Feynman diagrams representing the exclusive production of top pairs via photon-photon fusion at a lepton-hadron collider.

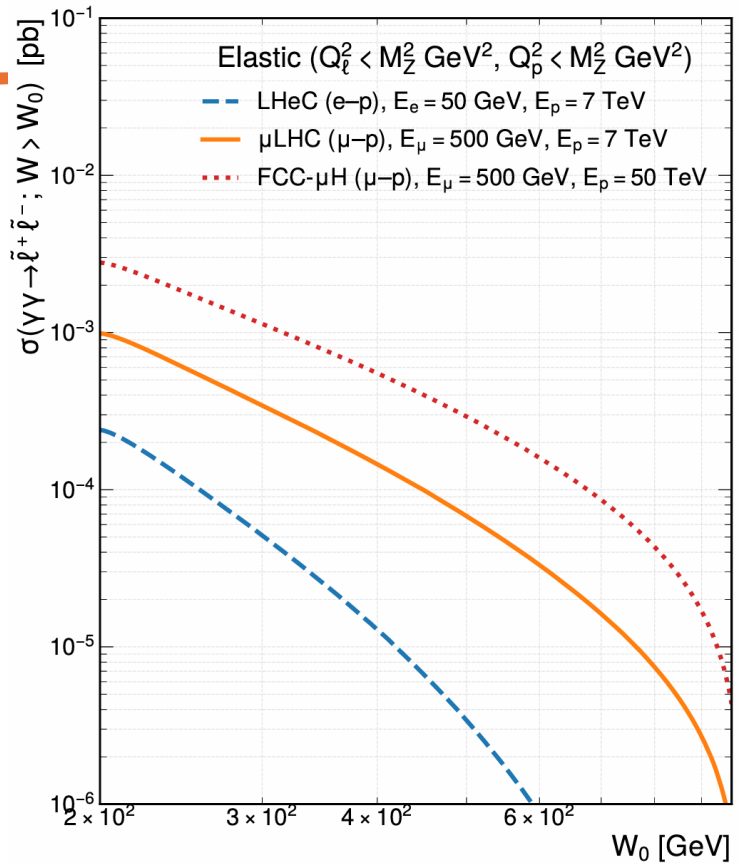
σ^{el}(LHeC@1.2 TeV)	= 0.046 fb
σ^{el}(μLHC@3.7 TeV)	= 0.486 fb
σ^{el}(FCC- μh@10 TeV)	= 1.832 fb

Elastic $t\bar{t}$ production cross sections (in pb) at the LHeC@1.2 TeV, LHμC@3.7 TeV & FCC- μh@10 TeV.

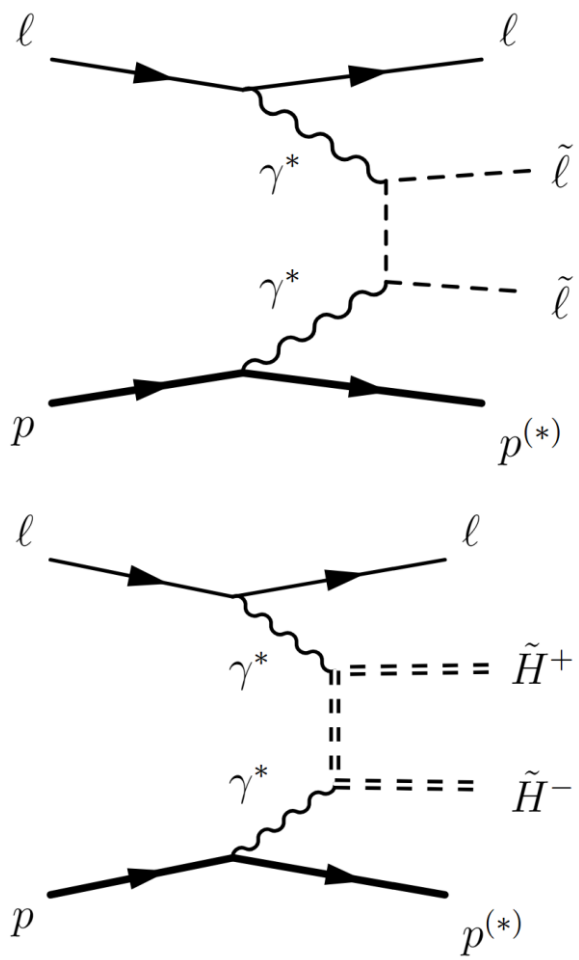


Two-photon exclusive production of supersymmetric pairs





Elastic production cross sections (in pb) of higgsinos and sleptons at the LHeC@1.2 TeV, LHμC@3.7 TeV & FCC- μh@10 TeV.



(top) exclusive pair production of scalar leptons (sleptons) & (bottom) two-photon production of higgsinos.

Anomalous triple/quartic gauge couplings

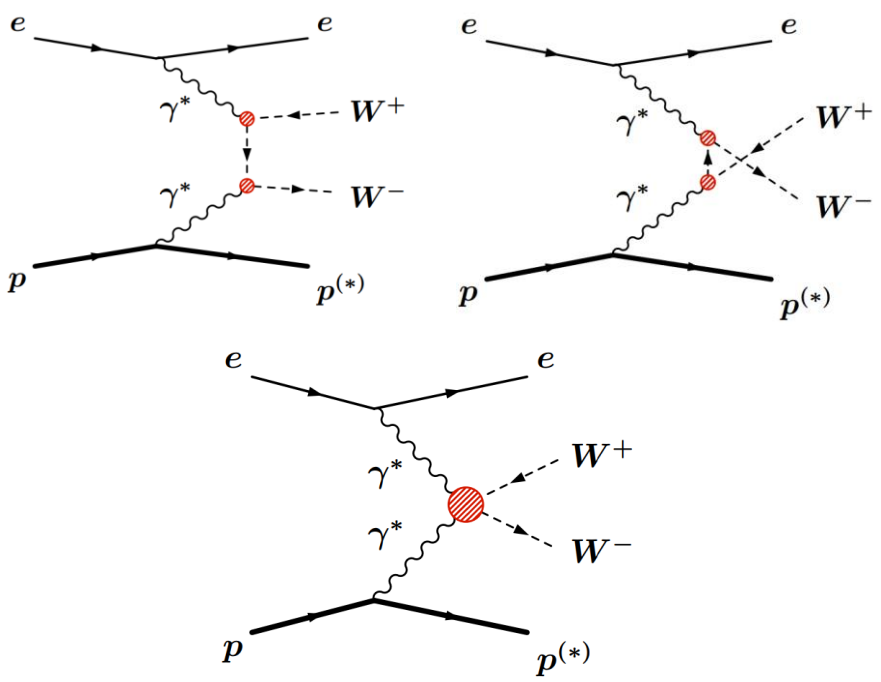
- In the EFT formalism, New Physics effects contributing to aQGCs are parameterized using higher-dimensional operators.
- The lowest-order modifications to quartic gauge couplings, which do not simultaneously induce triple gauge boson interactions, appear only atdimension-8.
- The general effective Lagrangian that governs these interactions is expressed as

$$\mathcal{L}_{\text{eff}} = \mathcal{L}_{\text{SM}} + \sum_i \frac{f_i}{\Lambda^4}\mathcal{O}_i$$

- The general form of the effective Lagrangian for anomalous quartic gauge couplings is given by,

$$\mathcal{L}_{\text{aQGC}} = \sum_{j=0}^{1} \frac{f_{S,j}}{\Lambda^4}\mathcal{O}_{S,j} + \sum_{j=0}^{7} \frac{f_{M,j}}{\Lambda^4}\mathcal{O}_{M,j} + \sum_{j=0}^{9} \frac{f_{T,j}}{\Lambda^4}\mathcal{O}_{T,j}.$$

CMS-PAS-SMP-24-019

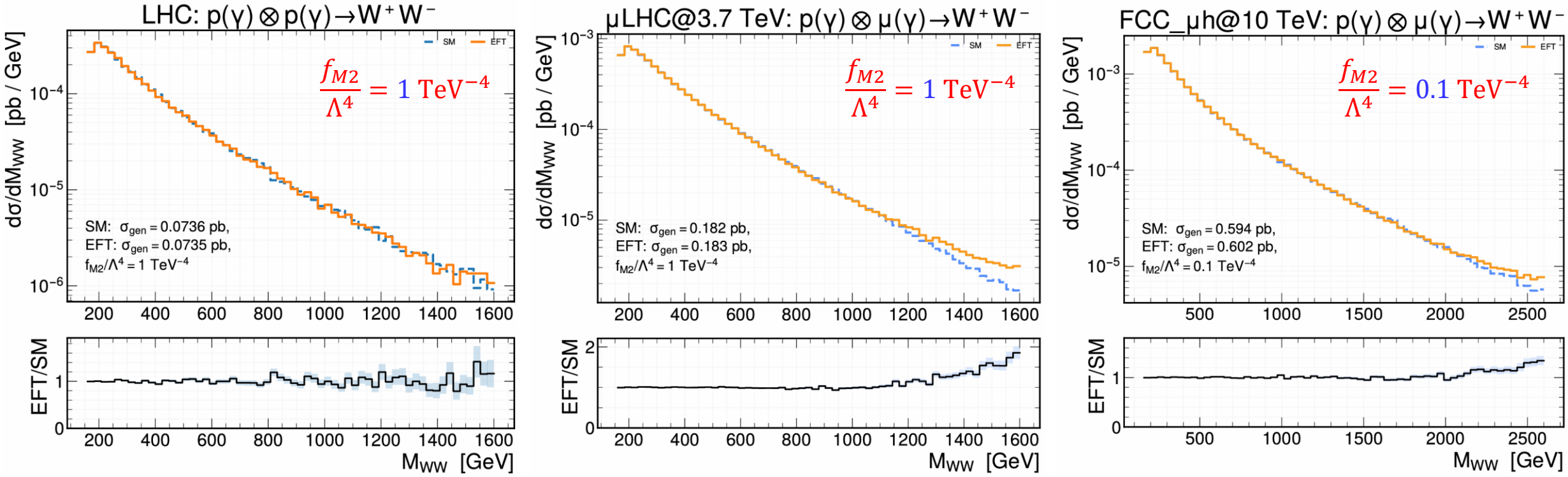


Clean observables: In $\gamma\gamma \to W^+W^-$, the aQGCs mainly boost the high-M_{WW} and high-p_T tails.

Elastic and dissociative $\gamma\gamma \to W^+W^-$ production in the clean electron-proton environment is a powerful test bench for anomalous triple and quartic gauge couplings.

$\gamma\gamma \to W^+W^-$ as an aQGC probe at three machines

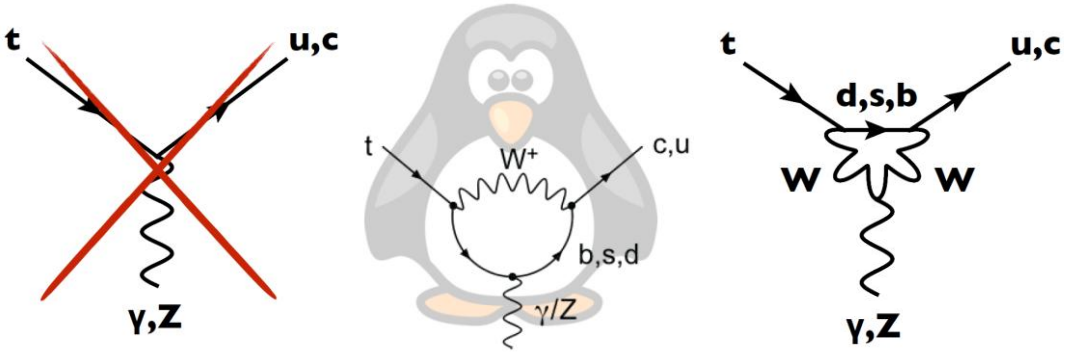
- Key pattern: EFT effects are tiny at low M_{WW} but grow rapidly in the high-mass tail, giving a clean shape handle on aQGCs.
- All machines highlights robust sensitivity to quartic gauge couplings; each facility contributes a different strength.
- μp : lowest backgrounds, precise kinematics, potential proton/muon tagging
- FCC-μh has the largest yield and hardest tail, hence the strongest statistical reach.



Generator-level M_{WW} spectra for LHC (pp), μLHC@3.7 TeV (μp) and FCC-μh@10 TeV (μp). The orange curve includes a dim-8 EFT, the blue curve is SM. Lower panels show EFT/SM.

The top quark FCNCs @ LHμC@3.7 TeV & FCC- μh@10 TeV

- The Flavor-changing neutral current (FCNC) process is forbidden at tree level in the SM and is highly suppressed in one-loop corrections by the Glashow-Iliopoulos-Maiani mechanism.
- The FCNC interactions of the tqX vertex (where *q* represents up and charm quarks) are key processes to search for New Physics.



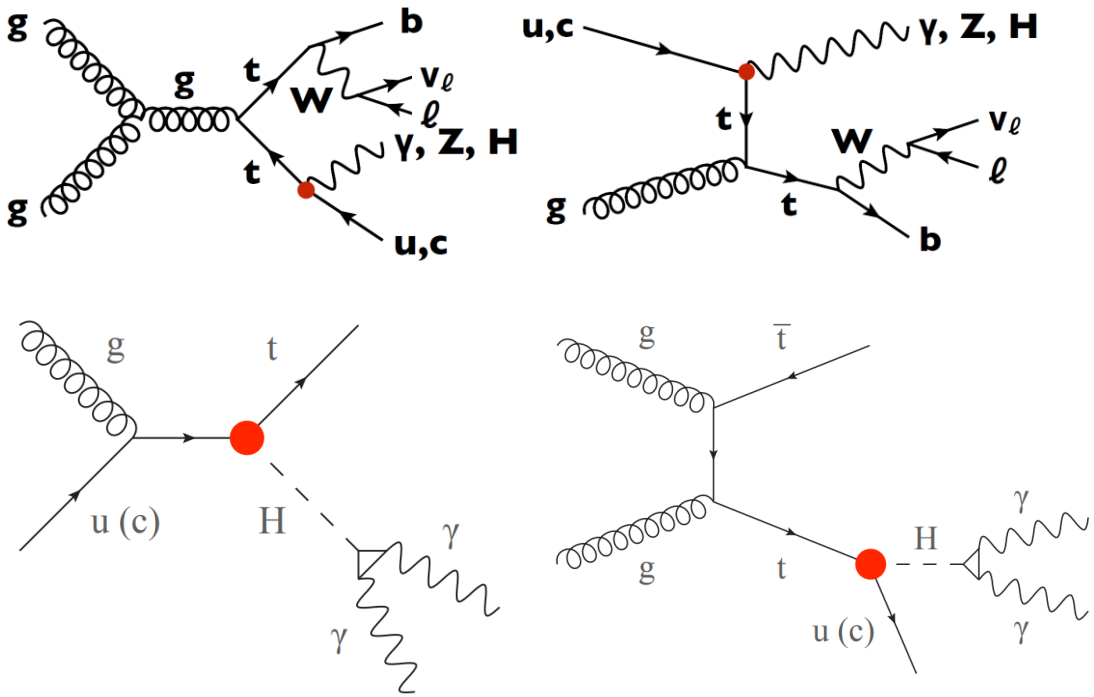
- The SM expected top quark FCNC transition is far from the sensitivity of the current experiments.
- The FCNC decay rates can be significantly enhanced in some New Physics models.
- The observation of a top quark FCNC interaction would signal the existence of New Physics.

	SM	QS	2HDM	FC 2HDM	MSSM	$\not{R}$ SUSY
$t \to uZ$	8×10^{-17}	1.1×10^{-4}	–	–	2×10^{-6}	3×10^{-5}
$t \to u\gamma$	3.7×10^{-16}	7.5×10^{-9}	–	–	2×10^{-6}	1×10^{-6}
$t \to ug$	3.7×10^{-14}	1.5×10^{-7}	–	–	8×10^{-5}	2×10^{-4}
$t \to uH$	2×10^{-17}	4.1×10^{-5}	5.5×10^{-6}	–	10^{-5}	$\sim 10^{-6}$
$t \to cZ$	1×10^{-14}	1.1×10^{-4}	$\sim 10^{-7}$	$\sim 10^{-10}$	2×10^{-6}	3×10^{-5}
$t \to c\gamma$	4.6×10^{-14}	7.5×10^{-9}	$\sim 10^{-6}$	$\sim 10^{-9}$	2×10^{-6}	1×10^{-6}
$t \to cg$	4.6×10^{-12}	1.5×10^{-7}	$\sim 10^{-4}$	$\sim 10^{-8}$	8×10^{-5}	2×10^{-4}
$t \to cH$	3×10^{-15}	4.1×10^{-5}	1.5×10^{-3}	$\sim 10^{-5}$	10^{-5}	$\sim 10^{-6}$

Aguilar-Saavedra J.A., Top flavor-changing neutral interactions: Theoretical expectations and experimental detection. *Acta Physica Polonica* B, 35 (2004) 2695.

Summary of ATLAS and CMS results

- Over the years, various experiments have searched for the FCNC processes through the anomalous production or decay of top quark.
- No clear evidence for the presence of top quark FCNC interactions is observed and upper limits are set on the anomalous couplings and top quark branching ratio.



Boson	Best Limit on the BR (95% CL)		
γ	ATLAS 81 fb^{-1}, focus on $\sigma_{t+boson}$	~0.3 x 10^{-4} (up)	~2 x 10^{-4} (charm)
Z	ATLAS 36 fb^{-1}, focus on BR	~2 x 10^{-4} (up)	~2 x 10^{-4} (charm)
Higgs	CMS 137 fb^{-1}, $H \rightarrow \gamma\gamma$	~2 x 10^{-4} (up)	~7 x 10^{-4} (charm)
gluon	CMS Run-1, top+jet production	~0.2 x 10^{-4} (up)	~4 x 10^{-4} (charm)

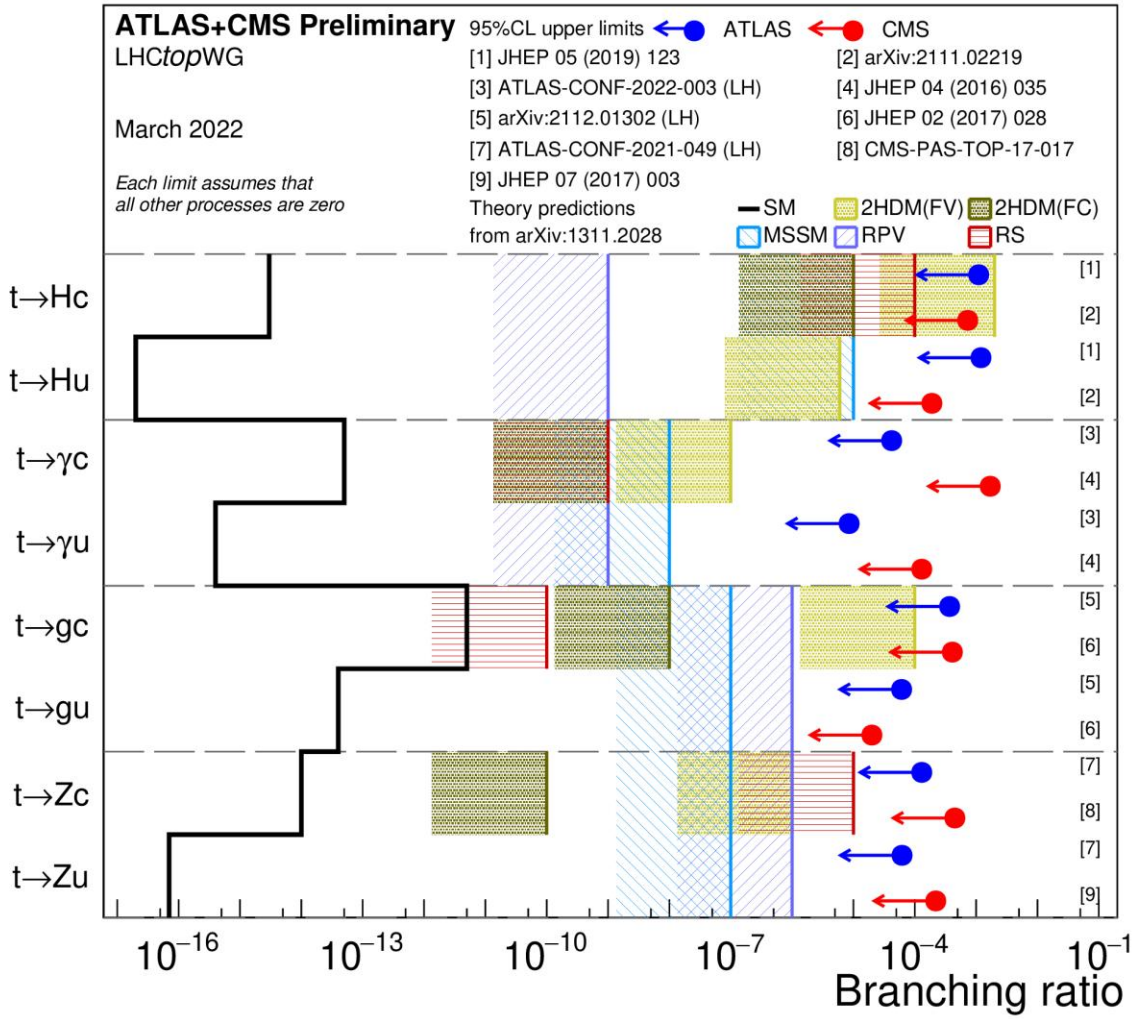
ATLAS Collaboration, Phys. Lett. B 800 (2020) 135082, 1908.08461 [hep-ex].
ATLAS Collaboration, ATLAS-CONF-2022-003 & ATLAS-CONF-2022-013.
ATLAS Collaboration, CMS-TOP-20-007, 2111.02219 [hep-ex].
ATLAS Collaboration, JHEP02 (2022) 169, 2112.09734 [hep-ex].

The SM & beyond the SM & Experimental results

- Several extensions of the SM predict much higher branching ratios up to 10^8 - 10^9 order of magnitude larger than SM predictions.

- Any observation of these rare FCNC transitions would be a clear signal of new physics beyond the SM.

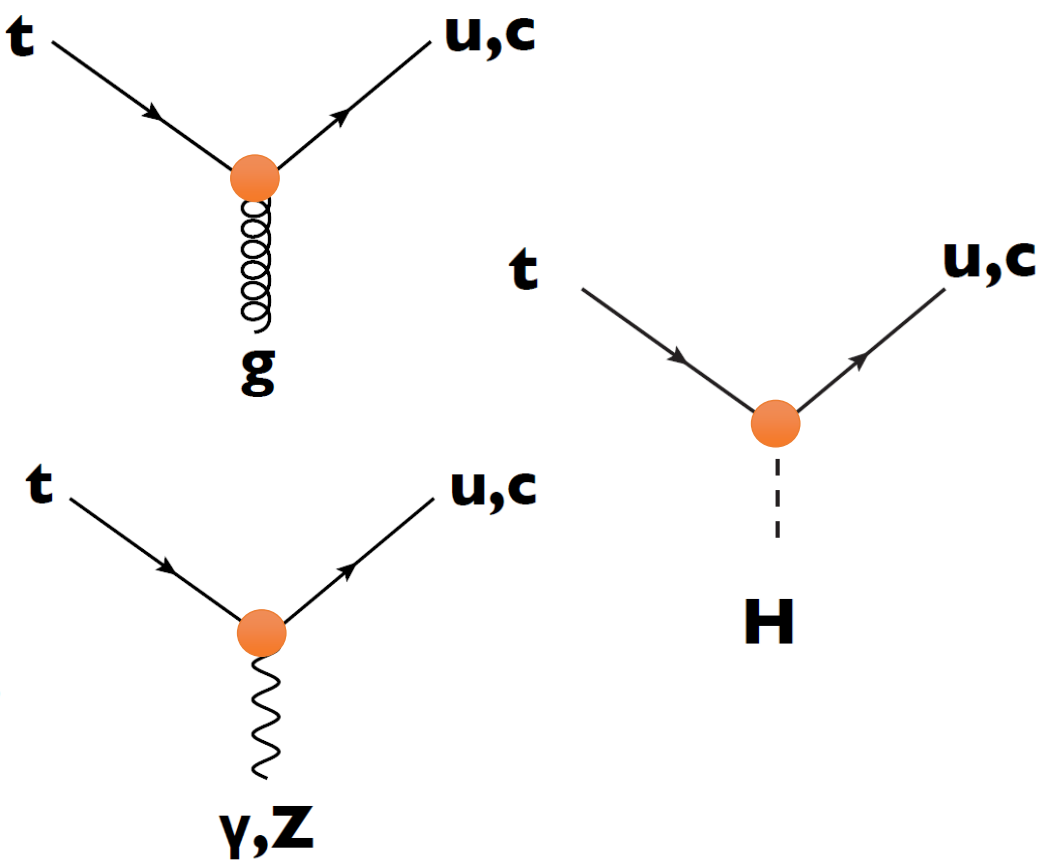
Aguilar-Saavedra J.A., Top flavor-changing neutral interactions: Theoretical expectations and experimental detection. *Acta Physica Polonica* B, 35 (2004) 2695.



The Effective Lagrangian Approach

- The most general effective Lagrangian describing the FCNC interactions can be parametrized as

$$
\begin{aligned}
\mathcal{L}_{\mathrm{FCNC}} = \sum_{q=u,c} \Bigg[& \frac{g_s}{2m_t} \bar{q}\lambda^a \sigma^{\mu\nu} (\zeta^L_{qt} P^L + \zeta^R_{qt} P^R) t G^a_{\mu\nu} \\
& - \frac{1}{\sqrt{2}} \bar{q} (\eta^L_{qt} P^L + \eta^R_{qt} P^R) t H \\
& - \frac{g_W}{2c_W} \bar{q}\gamma^\mu (X^L_{qt} P_L + X^R_{qt} P_R) t Z_\mu \\
& + \frac{g_W}{4c_W m_Z} \bar{q}\sigma^{\mu\nu} (\kappa^L_{qt} P_L + \kappa^R_{qt} P_R) t Z_{\mu\nu} \\
& + \frac{e}{2m_t} \bar{q}\sigma^{\mu\nu} (\lambda^L_{qt} P_L + \lambda^R_{qt} P_R) t A_{\mu\nu} \Bigg] + \mathrm{H.c.},
\end{aligned}
$$



https://feynrules.irmp.ucl.ac.be/wiki/EffectiveModels
https://feynrules.irmp.ucl.ac.be/wiki/GeneralFCNTop

- Top-quark FCNC interactions (tqγ, tqZ, tqH)
- Lepton–hadron colliders: μLHC and FCC-μH

- SMEFT & anomalous FCNC operators (dim-6)
- Event generation & detector modeling for μp

- High-energy PDFs (proton + photon/EPAs)
- MadGraph5_aMC@NLO,
- UFO model implementation, parameter scans

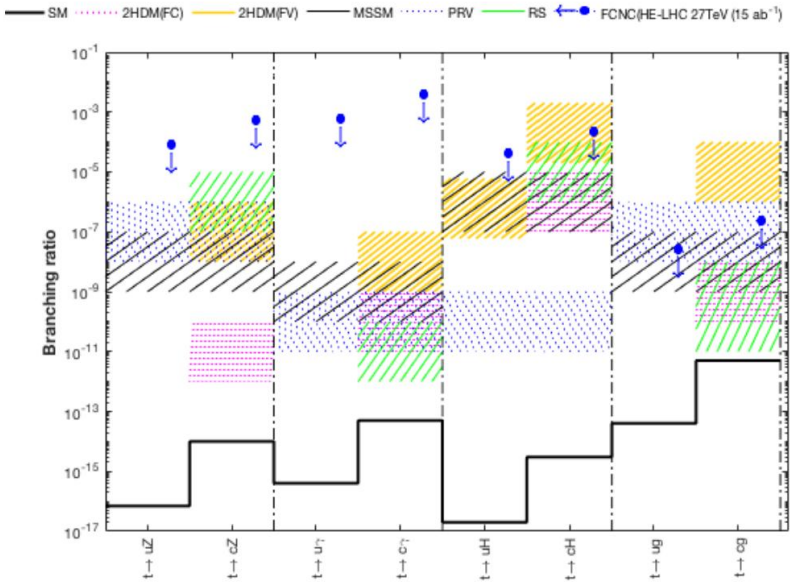
- Signal significance estimates and limit setting

- Top quark FCNC at HE-LHC and FCC-hh

  Hamzeh Khanpour, *Nucl. Phys. B* 958 (2020) 115141

- Top quark FCNC at FCCee

  Hamzeh Khanpour, Phys. Lett. B 775 (2017) 25-31



Collider Configurations and MG settings

- Top-FCNC EFT model: import model topFCNC_UFO

$$p \equiv \{g,\ u,\ c,\ d,\ s,\ b,\ \bar{u},\ \bar{c},\ \bar{d},\ \bar{s},\ \bar{b}\}$$

$$j \equiv \{g,\ u,\ c,\ d,\ s,\ \bar{u},\ \bar{c},\ \bar{d},\ \bar{s},\ b,\ \bar{b}\}$$

- generate p mu+ > mu+ t NP=1

$$p + \mu^+ \longrightarrow \mu^+ + t,$$

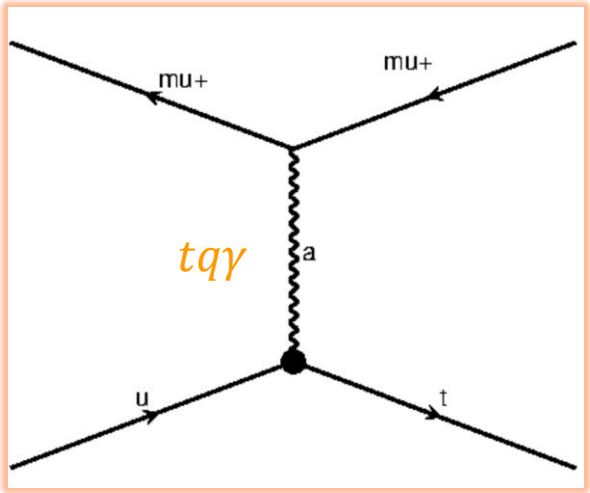
μLHC@3.7 TeV (μp) $\quad p(7000\ \mathrm{GeV}) + \mu^+(500\ \mathrm{GeV})$

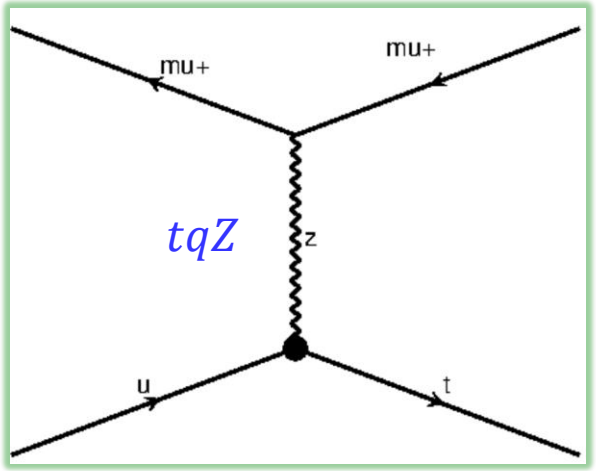
FCC-μh@10 TeV (μp) $\quad p(50000\ \mathrm{GeV}) + \mu^+(500\ \mathrm{GeV})$

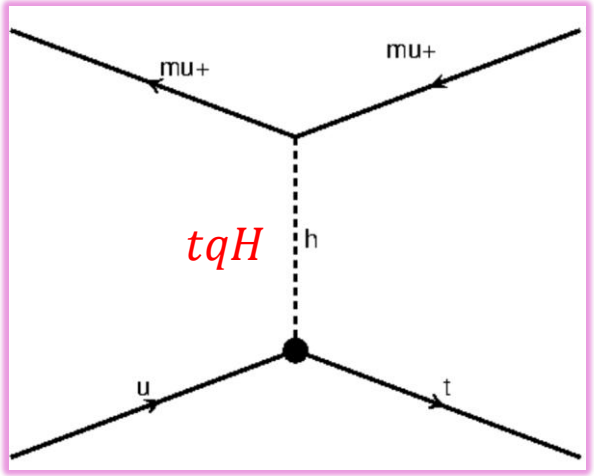
Generation cuts

$$p_T^\ell \geq 5\ \mathrm{GeV},$$
$$0 \leq |\eta^\ell| \leq 5,$$

Top-FCNC couplings: $tq\gamma$; tqZ; tqH





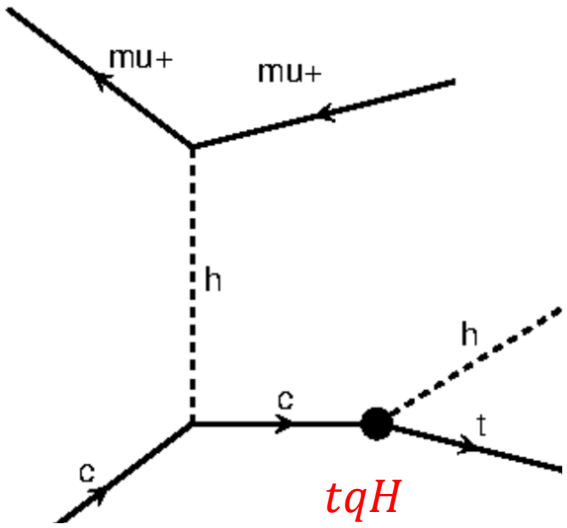


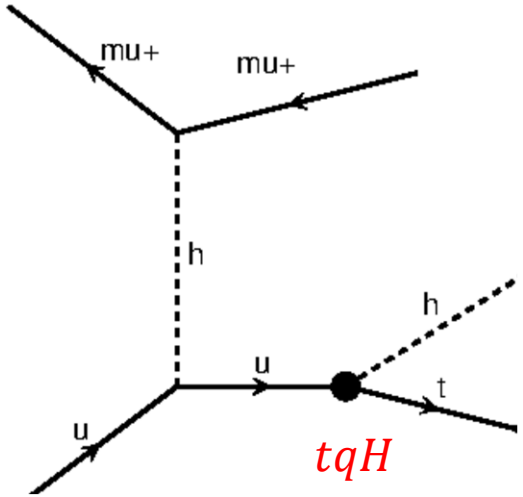
$$
\begin{aligned}
\mathcal{L}_{\mathrm{FCNC}} = \sum_{q=u,c} \Bigg[& \frac{g_s}{2m_t} \bar{q}\lambda^a \sigma^{\mu\nu} (\zeta^L_{qt} P^L + \zeta^R_{qt} P^R) t G^a_{\mu\nu} \\
& - \frac{1}{\sqrt{2}} \bar{q} (\eta^L_{qt} P^L + \eta^R_{qt} P^R) t H \\
& - \frac{g_W}{2c_W} \bar{q}\gamma^\mu (X^L_{qt} P_L + X^R_{qt} P_R) t Z_\mu \\
& + \frac{g_W}{4c_W m_Z} \bar{q}\sigma^{\mu\nu} (\kappa^L_{qt} P_L + \kappa^R_{qt} P_R) t Z_{\mu\nu} \\
& + \frac{e}{2m_t} \bar{q}\sigma^{\mu\nu} (\lambda^L_{qt} P_L + \lambda^R_{qt} P_R) t A_{\mu\nu} \Bigg] + \mathrm{H.c.},
\end{aligned}
$$

Top-FCNC couplings: tqH

generate p mu+ > mu+ t h j NP=1

$$p\,\mu^+ \to \mu^+\,t\,h\,j \qquad (\mathrm{FCNC},\ \mathrm{EFT},\ NP=1)$$





- All of these topics require separate, detailed studies - each with its own selections, systematics, and limit-setting - to properly quantify sensitivity across μLHC, and FCC-μH.

Thank
you!